

# SkySoft

## Internet Cafe Management System

---

### Internet Cafe Dashboard

Frontend design based on the console app menu and database structure.

18/04/2026, 17:13:46

Total Customers

**6**

Registered customers in the system

Active Sessions

**1**

Currently in progress

Available PCs

**3**

Ready for new sessions

Total Sessions

**5**

Stored in session history

#### Quick Actions

These match the main features in the console app.

Start SessionEnd SessionAdd Customer

#### Add Customer

Console option 10

| Customer ID                                    | Customer Name                                     |
|------------------------------------------------|---------------------------------------------------|
| <input type="text" value="e.g. C007"/>         | <input type="text" value="Enter full name"/>      |
| <input type="text" value="example@email.com"/> | <input type="text" value="Starting Balance (£)"/> |
|                                                | <input type="text" value="15.00"/>                |

#### Top Up Customer Balance

Console option 9

Select Customer

C001 - Hasan

Top Up Amount (£)

## Introduction

The SkySoft Internet Cafe Management System is a C# .NET console-based application developed to support the efficient operation of an internet cafe. The system is designed to manage key aspects of the business, including customer records, PC availability, and session tracking. By integrating a SQL Server database, the application enables persistent storage and retrieval of data such as customer details, session history, and account balances. Core functionalities include starting and ending user sessions, searching for customers, monitoring active sessions, and managing account top-ups. The system also demonstrates the implementation of custom data structures, such as hash tables and linked lists, to efficiently handle data operations and reinforce fundamental programming concepts.

This report is structured to present the development and evaluation of the system in a clear and logical manner. The design section outlines the architecture of the application, including system

---

---

structure, database design, and key implementation decisions. The testing section describes the methods used to verify functionality and ensure reliability. The conclusion summarises the overall outcomes of the project, followed by a Summary of key findings. The report then discusses limitations & Issues encountered during development and suggests possible improvements. Finally, the References section lists all sources used throughout the project.

## **Design**

### **Front-End**

A web-based front-end prototype was developed using HTML, CSS and JavaScript. This interface mirrors the features of the console system while improving usability through a visual dashboard.

#### **Main Features Included:**

- Dashboard statistics
- Add customer form
- Start session
- End session
- Search customer by ID or name
- View all sessions
- View active sessions
- View PC availability
- View session history
- Top up customer balance

#### **User Interface Design Principles**

The front-end uses:

- Responsive grid layout
- Navigation sidebar
- Cards for dashboard statistics
- Tables for records
- Real-time clock

- 
- Colour-coded status badges
  - Form validation feedback

This creates a professional and user-friendly interface.

## Back-End

The design of the SkySoft Internet Café Management System follows a modular and structured approach, separating concerns across different layers to improve maintainability and scalability. The application is implemented as a C# .NET console program, with a clear division between data access, business logic, and user interaction. A SQL Server database is used as the backend to store persistent data, including customers, PCs, and session records, while a dedicated database helper component manages all communication between the application and the database. The system architecture also incorporates custom-built data structures, such as a hash table for efficient customer lookup and linked lists for managing PCs and session data, ensuring both performance and practical demonstration of core data structure concepts.

```
InternetCafeManagementSystem/
├── Data/
│   └── DatabaseHelper.cs      - SQL database operations
├── DataStructures/
│   ├── CustomHashTable.cs    - Custom hash table implementation
│   └── CustomLinkedList.cs    - Custom linked list implementation
├── Models/
│   ├── Customer.cs           - Customer model
│   ├── PC.cs                 - PC model
│   └── Session.cs            - Session model
├── Services/
│   └── InternetCafeService.cs - Business logic layer
└── Program.cs                - Entry point and menu
```

## Models

Three primary classes were used:

### Customer

- Customer ID

- 
- Name
  - Email
  - Balance

## **PC**

- PC ID
- Availability status

## **Session**

- Session ID
- Customer ID
- PC ID
- Start time
- End time
- Cost

# **Key Functional Processes**

## **Add Customer**

The system validates:

- Required fields
- Unique customer ID
- Numeric balances
- Restricted/offensive text filtering

## **Start Session**

When a session begins:

1. Available PC selected
2. New session ID generated
3. Session stored
4. PC marked occupied

---

## End Session

When ending a session:

1. Session selected
2. End time recorded
3. Cost calculated
4. PC returned to available

## Search Functions

Customers can be found by:

- Customer ID
- Name

## Testing

### Functional Testing

Functional testing was used to confirm that the application features met their expected requirements.

| Test Case ID | Test Description                    | Expected Result             | Actual Result                          | Status |
|--------------|-------------------------------------|-----------------------------|----------------------------------------|--------|
| TC01         | Startup without database configured | Clear setup warning shown   | Application crashed with SQL exception | Fail   |
| TC02         | Startup after database setup        | Main menu displayed         | Main menu displayed successfully       | Pass   |
| TC03         | Search customer by ID (C001)        | Existing customer displayed | Hasan displayed correctly              | Pass   |

---

|      |                                        |                                |                                        |      |
|------|----------------------------------------|--------------------------------|----------------------------------------|------|
| TC04 | Search non-existing customer ID (C009) | Customer not found message     | Correct message displayed              | Pass |
| TC05 | Search customer by name                | Matching customer displayed    | Hasan displayed correctly              | Pass |
| TC06 | View PC availability                   | PC list displayed              | PC01, PC02, PC03 displayed             | Pass |
| TC07 | Top up balance                         | Balance increases successfully | Balance updated successfully           | Pass |
| TC08 | Invalid menu input ("abc")             | Error message, no crash        | Invalid option shown, continued safely | Pass |

## Unit Testing

Automated unit tests were carried out using MSTest to verify the hash table.

| Test Case ID | Unit Test      | Expected Result               | Actual Result | Status |
|--------------|----------------|-------------------------------|---------------|--------|
| UT01         | Add valid item | Item inserted successfully    | Test passed   | Pass   |
| UT02         | Retrieve item  | Correct stored value returned | Test passed   | Pass   |
| UT03         | Duplicate key  | Duplicate key rejected        | Test passed   | Pass   |

---

## Limitations & Issues Encountered

During development, several limitations were identified:

1. Synchronising front-end and back-end systems required more time.
2. The prototype front-end uses temporary JavaScript arrays rather than permanent storage.
3. Error handling could be expanded further.
4. No login/authentication system was implemented.
5. Scalability for large commercial cafes was not tested.

## Future Developments

The system could be improved through:

1. Full API connection between front-end and C# backend
2. Staff/admin login accounts
3. Barcode or membership card scanning
4. Booking/reservation system
5. Mobile companion app

## Conclusion

In conclusion, the SkySoft Internet Cafe Management System successfully demonstrates the development of a structured business management application using C# .NET, SQL Server, HTML, CSS and JavaScript. The system solves key operational tasks including customer management, session tracking and PC monitoring.

The use of both database technologies and custom-built data structures highlights strong technical understanding. Furthermore, the front-end prototype shows clear awareness of usability and modern interface design.

Overall, the project provides a strong foundation for future commercial development and demonstrates practical programming, testing and software engineering skills.